'Tackling' rugby safety through a collective approach

Sharief Hendricks ,[1,2] Carolyn Emery ,[3,4,5,6,7,8,9,10] Ben Jones ,[1,2,11,12,13] James Craig Brown ,[1,2,14] Kathryn Dane ,[15] Stephen W West,[3,16,17] Keith A Stokes,[16,17,18] Richie Gray,[19] Ross Tucker[14,20]

COMMITMENT TO SEEKING AN EVIDENCE-INFORMED APPROACH

When it comes to player welfare, Rugby Union governing bodies have committed to seeking and funding an evidence-informed approach. This involves using research to make informed decisions about policy, laws and injury prevention programmes. Over the last decade, a growing body of research has informed player safety, for example, modifications to scrum laws to reduce catastrophic head and neck injuries.[1] However, major gaps remain, including tackle research focusing on the women's game.[2]

A SOCIOECOLOGICAL PERSPECTIVE

It is well understood that player welfare, specifically injury prevention, is a complex issue. To effectively address these complexities and make a long-term impact requires a dynamic socioecological approach.[3 4] An athlete operates within a socioecological structure (individual, interpersonal, organisational, community) that is influenced by a web of inter-related factors and actors, both of which change over time and/or when a factor/actor is modified (figure 1). Typically, injury prevention research will identify player level factors that influence injury risk (risk factors), and aim to modify these factors through behaviour change interventions. However, the socioecological view emphasises understanding contextual factors influencing implementation of such modifications. For example, if the behaviour intervention is a training programme, how much time is available to implement the training programme? Or, is the training programme appropriate for all sexes?

REDUCING CONCUSSION RISK DURING THE TACKLE

Reducing concussion risk during the tackle, at all levels, is a top priority for rugby stakeholders and governing bodies. Video analyses examining head impacts and resulting concussions have identified that elements of players' tackling technique are major risk factors. For both effective and safe tackling, the optimal techniques are to maintain a low body position or dip from an upright to low body position to contact the ball-carrier at their centre of gravity (mid-torso area). Tackles causing head injuries differ, with upright tackler body positions and head-to-head and head-to-shoulder contact found to be over-represented in injurious tackles.[5]

In view of the concussion risk associated with direct head contacts caused in part by an upright body position,[5] World Rugby (WR) has applied a regulatory approach to change players' tackling behaviour through more strict and consistent refereeing of high tackles.[6] WR and national governing bodies have also trialled lowering the height of a legal tackle from the shoulder to the armpit in an effort to 'nudge' players to make contact with the ball-carrier mid-torso.[7 8] France, New Zealand and England have also announced they will implement lowering the legal tackle height in community rugby. These tackle law changes are related to the maximum height of a legal tackle for officiating purposes, and reinforce the optimal techniques recommended for a safe and effective tackle, that is, contacting the ball-carrier at their centre of gravity.

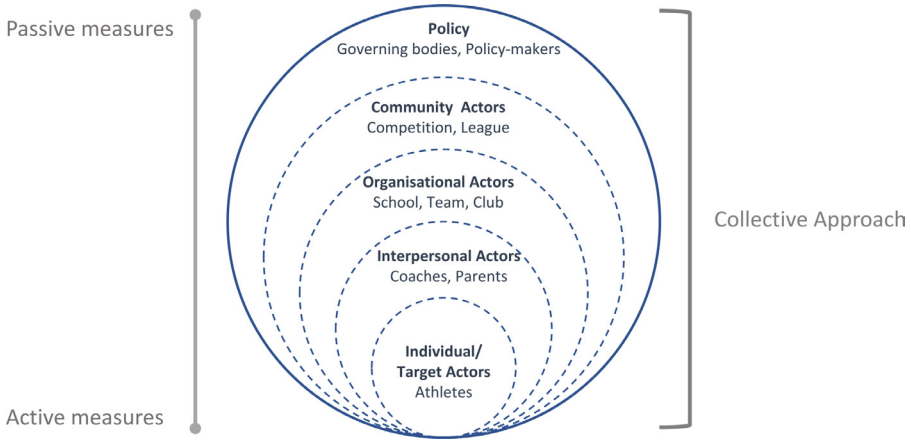

Figure 1 The collective approach of the different actors in the socioecological view for injury prevention in sport along the passive–active intervention continuum. Adapted from [3 4 9]

[1]Division of Physiological Sciences and Health through Physical Activity, Lifestyle and Sport Research Centre, Department of Human Biology, Faculty of Health Sciences, University of Cape Town, Cape Town, Western Cape, South Africa
[2]Carnegie Applied Rugby Research (CARR) Centre, Carnegie School of Sport, Leeds Beckett University, Leeds, UK
[3]Sport Injury Prevention Research Centre, Faculty of Kinesiology, University of Calgary, Calgary, Alberta, Canada
[4]Integrated Concussion Research Program, University of Calgary, Calgary, Alberta, Canada
[5]Alberta Children's Hospital Research Institute, University of Calgary, Calgary, Alberta, Canada
[6]Hotchkiss Brain Institute, University of Calgary, Calgary, Alberta, Canada
[7]O'Brien Institute for Public Health, University of Calgary, Calgary, Alberta, Canada
[8]Departments of Pediatrics, Cumming School of Medicine, Calgary, Alberta, Canada
[9]Community Health Sciences, Cumming School of Medicine, University of Calgary, Calgary, Alberta, Canada
[10]McCaig Institute for Bone and Joint Health, University of Calgary, Calgary, Alberta, Canada
[11]Premiership Rugby, London, UK
[12]England Performance Unit, Rugby Football League, Manchester, UK
[13]School of Behavioural and Health Sciences, Faculty of Health Sciences, Australian Catholic University, Brisbane, QLD, Australia
[14]Institute of Sport and Exercise Medicine, Department of Exercise, Sport and Lifestyle Medicine, Faculty of Medicine and Health Sciences, Stellenbosch University, Stellenbosch, South Africa
[15]Discipline of Physiotherapy, School of Medicine, Trinity College Dublin, Dublin, Ireland
[16]Centre for Health, and Injury & Illness Prevention in Sport, University of Bath, Bath, UK
[17]UK Collaborating Centre on Injury and Illness Prevention in Sport (UKCCIIS), University of Bath, Bath, UK
[18]Rugby Football Union, London, UK
[19]GSI Performance Scottish Borders, Edinburgh, UK
[20]World Rugby, Dublin, Ireland

Correspondence to Dr Sharief Hendricks, Division of Physiological Sciences and Health through Physical Activity, Lifestyle and Sport Research Centre, Department of Human Biology, Faculty of Health Sciences, University of Cape Town, Cape Town, Western Cape, South Africa; sharief.hendricks01@gmail.com

Br J Sports Med: first published as 10.1136/bjsports-2023-107020 on 12 April 2023. Downloaded from http://bjsm.bmj.com/ on April 13, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.

Therefore, the change in the maximum height for a legal tackle strengthens our current coaching of proper tackling technique.

Changing laws and policies is a 'passive' measure on the passive–active injury prevention and intervention continuum.[9] In general, passive measures have a broader impact as they relate to the whole sport. In contrast, active measures require players, or any party directly responsible for player's safety, to deliberately be involved in protecting the player. Being actively involved also enhances the individual's ability to perform the desired behaviour change, that is, self-efficacy, which is a key determinant of behaviour change.[10] Integrating passive and active behaviour change measures is thought to be most effective in sustainable injury prevention.[9]

Upskilling to play with the laws

To gain self-efficacy in reducing concussion risk in tackles, players must have the ability to repeatedly meet the recognised safety requirement of contacting the ball-carrier at the mid-torso, a skill that requires physical, technical, tactical and psychological proficiency and capacity. A large body of literature describes how to optimise training for skill learning, maintenance, transfer and capacity building. This literature has only recently been applied to tackle training.[11] The interpersonal actor in the socioecological view of preventing injury prevention in team sports is the coach, the target of behaviour change for national injury prevention programmes such as BokSmart.[12] While research on the specific elements of how to best test, design, provide feedback and evaluate the tackle during training is emerging, we need to acknowledge the complexities of coaching, and recognise that the main outcome measure of a player's tackle ability is performance. Thus, any measure (active or passive) to reduce concussion risk must consider the impact on the player's tackle performance, and accept that any safety measure that impairs performance is unlikely to be implemented.

COLLECTIVE APPROACH

Further, we must appreciate that research alone may be unable to meet the urgent time frames for optimising tackle safety through skill training. Also, the effectiveness and sustainability of a safe tackle-skill programme may be reliant on elements of coaching ('the art of coaching') that cannot necessarily be assessed using a traditional research design. In rugby, and specifically for tackle safety, the need for engagement, collaboration and collective action between stakeholders at all socioecological levels has never been greater—this includes players, coaches, parents, clubs, administrators, governing bodies, media and match officials (figure 1). The necessary next step in this collective action towards improving tackle safety is to bring these stakeholders, along with researchers and clinicians, together to create a specialist tackle working group. The mandate of this working group would be to identify the most important focus areas and research questions to inform best practice and policy, and then manage and oversee the research process. A key part of this mandate will be to ensure that all research is effectively translated and disseminated to the wider rugby community.

In conclusion, reducing concussion risk during the tackle is complex and requires a dynamic socioecological perspective to integrate active and passive measures to change player behaviour. The behaviour change required to safely and effectively contest the tackle is a skill, and multistakeholder engagement, collaboration and collective action between players, coaches, parents, clubs, administrators, governing bodies, media and match officials is needed to effectively reduce the risk of concussion in the tackle.

Twitter Sharief Hendricks @Sharief_H, Carolyn Emery @CarolynAEmery, Ben Jones @23benjones, James Craig Brown @jamesbrown06, Kathryn Dane @kathryndane2, Stephen W West @westy1609, Keith A Stokes @drkeithstokes, Richie Gray @RichieGrayGSI and Ross Tucker @Scienceofsport

Acknowledgements The authors would like to acknowledge Demi Davidow, Steve den Hollander, Isla Shill, Emily Spiegelhalter, Matt Hancock, Riaan van Tonder.

Contributors SH drafted the manuscript, CE, BJ, JCB, KD, SWW, KAS, RG, RT provided input. All authors edited and reviewed the final version.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests BJ is employed in a consultancy capacity by Premiership Rugby, the Rugby Football League and Leeds Rhinos Rugby League club.KS is employed by the Rugby Football Union, the national governing body for rugby union in England. RT is a research consultant for World Rugby. SH is a member of the BJSM editorial board.

Patient consent for publication Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

© Author(s) (or their employer(s)) 2023. No commercial re-use. See rights and permissions. Published by BMJ.

Check for updates

To cite Hendricks S, Emery C, Jones B, *et al*. *Br J Sports Med* Epub ahead of print: [*please include* Day Month Year]. doi:10.1136/bjsports-2023-107020

Accepted 30 March 2023

Br J Sports Med 2023;**0**:1–2. doi:10.1136/bjsports-2023-107020

ORCID iDs
Sharief Hendricks http://orcid.org/0000-0002-3416-6266
Carolyn Emery http://orcid.org/0000-0002-9499-6691
Ben Jones http://orcid.org/0000-0002-4274-6236
James Craig Brown http://orcid.org/0000-0002-7778-7783
Kathryn Dane http://orcid.org/0000-0001-8083-6278

REFERENCES

1 Hendricks S, Lambert MI, Brown JC, *et al*. An evidence-driven approach to scrum law modifications in amateur rugby played in South Africa. *Br J Sports Med* 2014;48:1115–9.
2 Dane K, Simms C, Hendricks S, *et al*. Physical and technical demands and preparatory strategies in female field collision sports: a scoping review. *Int J Sports Med* 2022;43:1173–82.
3 Bogardus RL, Martin RJ, Richman AR, *et al*. Applying the socio-ecological model to barriers to implementation of ACL injury prevention programs: a systematic review. *J Sport Health Sci* 2019;8:8–16.
4 Keats MR, Emery CA, Finch CF. Are we having fun yet? fostering adherence to injury preventive exercise recommendations in young athletes. *Sport Med* 2012;42:175–84.
5 Tucker R, Raftery M, Kemp S, *et al*. Risk factors for head injury events in professional rugby Union: a video analysis of 464 head injury events to inform proposed injury prevention strategies. *Br J Sports Med* 2017;51:1152–7.
6 Raftery M, Tucker R, Falvey ÉC. Getting tough on concussion: how welfare-driven law change may improve player safety-a rugby union experience. *Br J Sports Med* 2020;55:527–9.
7 van Tonder R, Starling L, Surmon S, *et al*. Tackling sport-related concussion: effectiveness of lowering the maximum legal height of the tackle in amateur male rugby-a cross-sectional analytical study. *Inj Prev* 2023;29:56–61.
8 Stokes KA, Locke D, Roberts S, *et al*. Does reducing the height of the tackle through law change in elite men's rugby Union (the Championship, England) reduce the incidence of concussion? A controlled study in 126 games. *Br J Sports Med* 2021;55:220–5.
9 Verhagen E, van Mechelen W. *Sports injury research*. Oxford University Press, 2010.
10 Owoeye OBA, McKay CD, RÄisÄnen AM, *et al*. Psychosocial factors and the effects of a structured injury prevention workshop on coaches' self-efficacy to implement the 11+ exercise program. *Int J Exerc Sci* 2020;13:1459–75.
11 Hendricks S, Till K, Oliver JL, *et al*. Technical skill training framework and skill load measurements for the rugby union tackle. *Strength & Conditioning Journal* 2018;40:44–59.
12 Viljoen W, Patricios J. BokSmart-implementing a national rugby safety programme. *Br J Sports Med* 2012;46:692–3.

Br J Sports Med: first published as 10.1136/bjsports-2023-107020 on 12 April 2023. Downloaded from http://bjsm.bmj.com/ on April 13, 2023 at Please Use: 502999. Hea Health Science Library Rm 324. Protected by copyright.